

Predictive Analytics

Shrinkage Methods

OLS · Ridge · LASSO · Elastic Net

Body Fat Dataset

- 252 male subjects · 15 variables
- Response: Body Fat percentage
- 13 anthropometric predictors
- Benchmark: OLS vs Shrinkage methods

Dataset Overview

252

Observations

15

Variables

13

Predictors

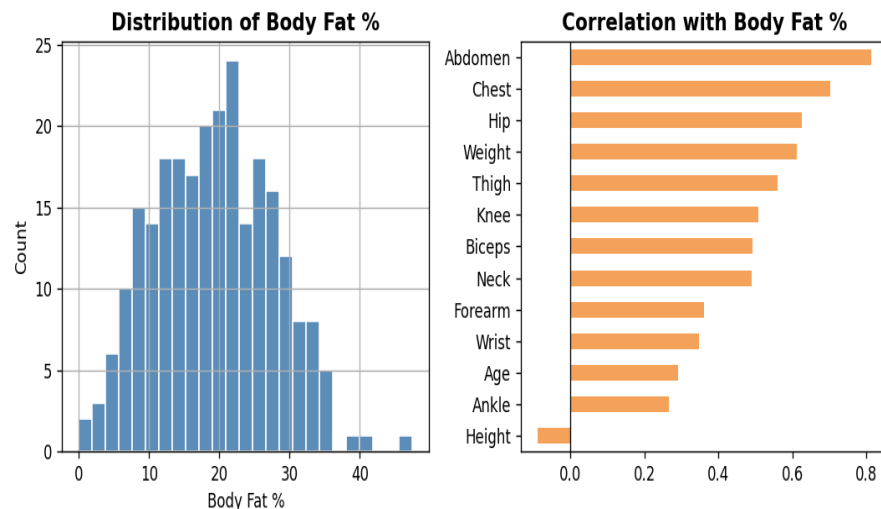
0

Missing Values

Key Variables

Variable	Description
BodyFat	Response – body fat percentage (%)
Abdomen	Abdominal circumference (cm)
Weight	Body weight (lbs)
Age	Age in years
Forearm	Forearm circumference (cm)
Wrist	Wrist circumference (cm)

EDA: Correlation with Body Fat



Abdomen shows the strongest positive correlation with body fat ($r \approx 0.81$)

Methodology

1 Data Split

152 training · 100 test observations
Random split (seed=42)

2 Standardisation

All predictors scaled to mean=0, std=1
Essential for fair penalisation

3 OLS Baseline

Ordinary Least Squares — no penalty
Establishes benchmark MSE

4 Ridge (L2)

Shrinks coefficients toward zero
 λ selected by 10-fold CV

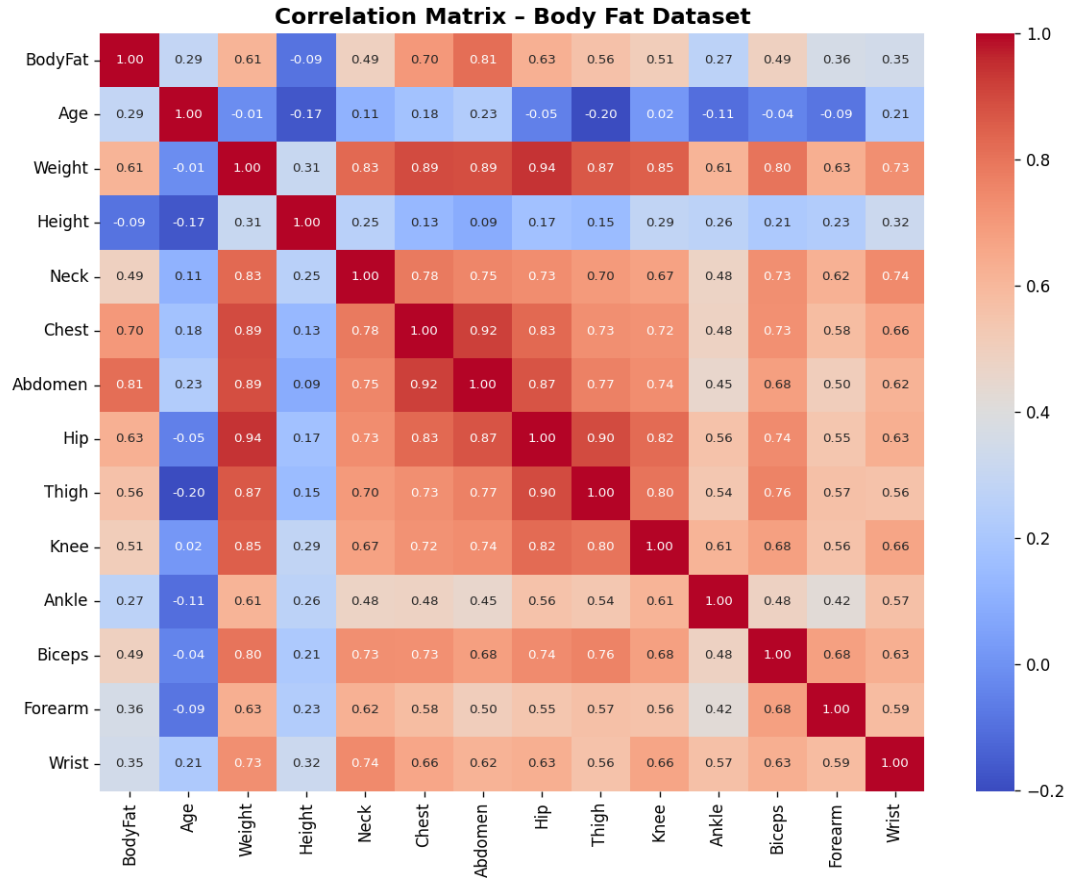
5 LASSO (L1)

Sets some coefficients exactly to zero
Automatic variable selection

6 Elastic Net

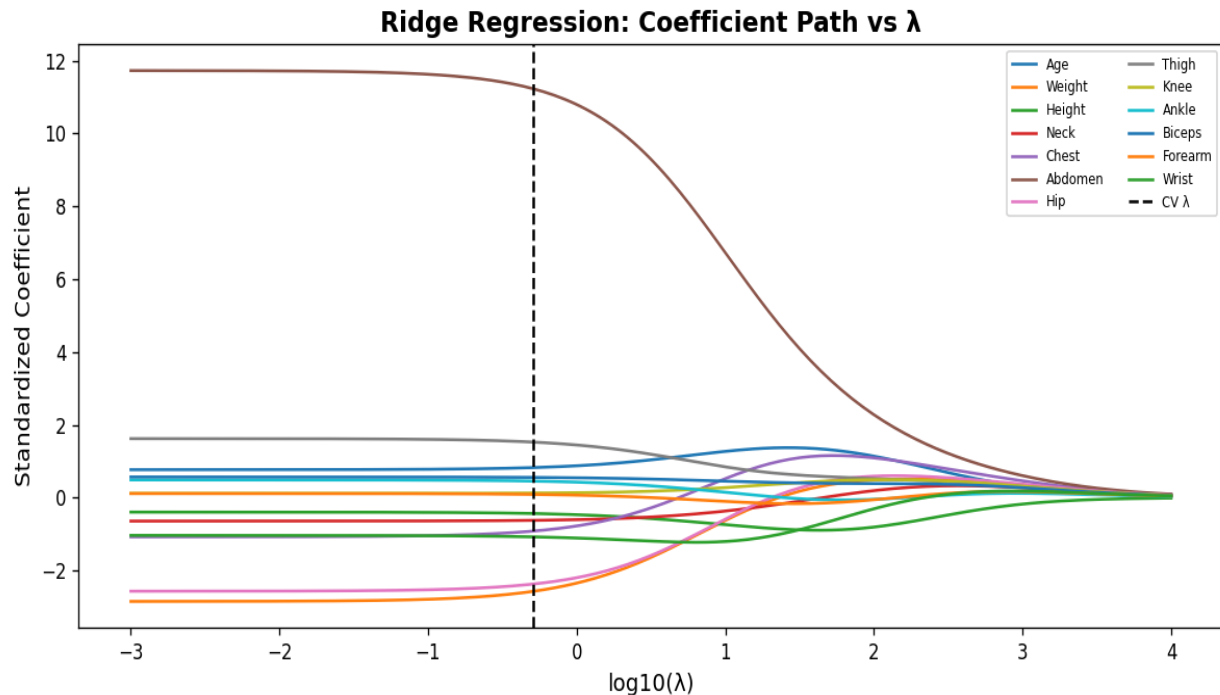
Combines Ridge (L2) + LASSO (L1)
Searches over α and l1_ratio by CV

Multicollinearity: Correlation Heatmap



- Chest, Abdomen, Hip & Weight are highly intercorrelated ($r > 0.8$)
- Abdomen has the strongest link to BodyFat ($r \approx 0.81$)
- High multicollinearity inflates OLS coefficient variance
- Ridge/LASSO regularisation handles this effectively

Ridge Regression – Coefficient Path



0.511

Optimal λ

20.98

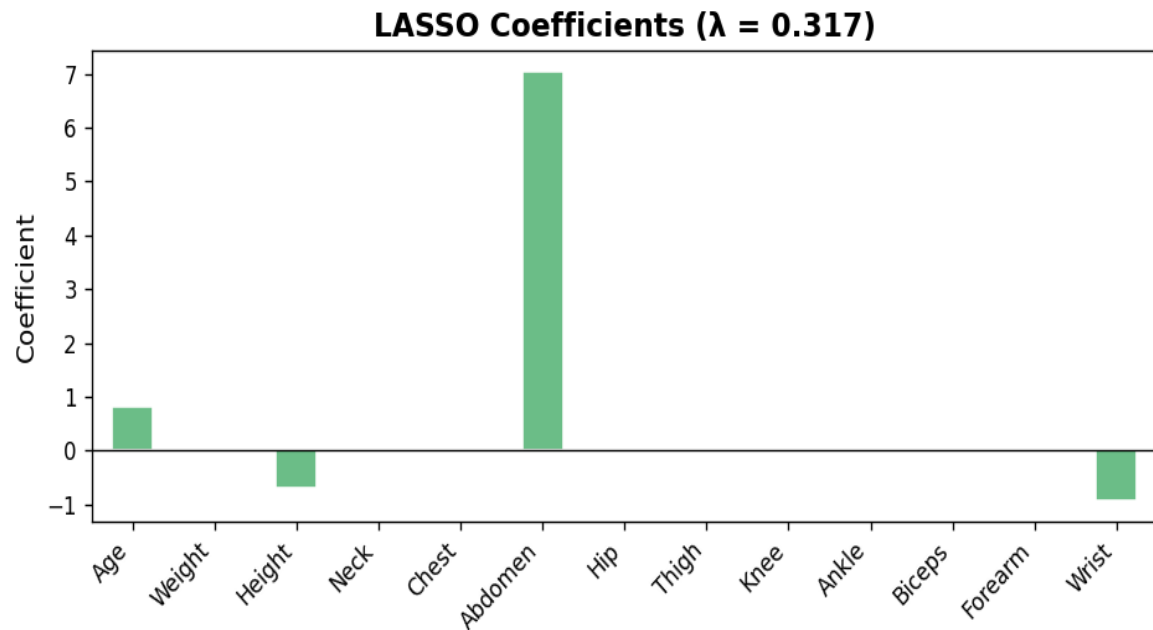
Test MSE

13 / 13

Predictors

Key Insight: As λ increases, all coefficients shrink toward zero but never reach exactly zero. Ridge retains all predictors — useful when every feature contributes.

LASSO – Variable Selection



LASSO sets 9 of 13 coefficients to exactly zero → sparse, interpretable model

Selected Predictors ($\lambda = 0.317$)

✓ Age

✓ Abdomen

✓ Forearm

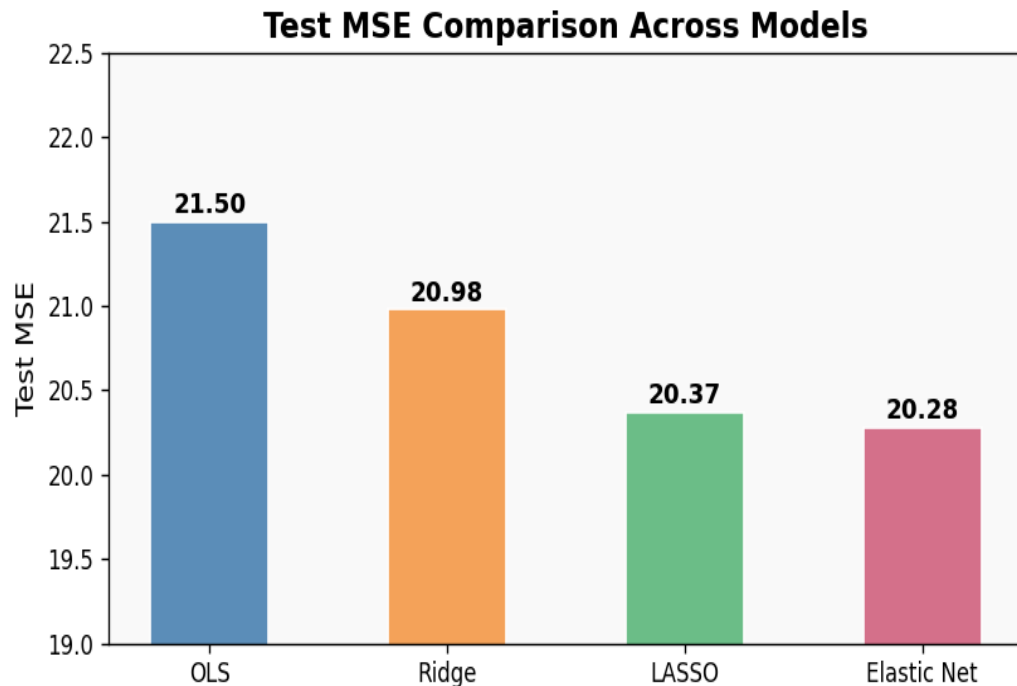
✓ Wrist

Optimal λ : 0.317

Test MSE: 20.37

Non-zero: 4 / 13

Model Comparison – Test MSE



Model	Test MSE	Non-zero
OLS	21.50	13
Ridge	20.98	13
LASSO	20.37	4
Elastic Net	20.28 ★	4

★ Best model: lowest test MSE

All shrinkage methods outperform plain OLS on unseen data

Elastic Net

Ridge (L2)

$\lambda = 0.511$
 $l1_ratio = 0$

No variable
selection

All 13
predictors

Test MSE:
20.98

LASSO (L1)

$\lambda = 0.317$
 $l1_ratio = 1$

Variable
selection

4 / 13
predictors

Test MSE:
20.37

Elastic Net

$\lambda = 0.267$
 $l1_ratio = 0.9$

Variable
selection

4 / 13
predictors

Test MSE:
20.28 ★

Conclusions

OLS overfit due to multicollinearity — Test MSE of 21.50 was the highest

Ridge (MSE 20.98) improved over OLS by shrinking coefficients but kept all 13 predictors

LASSO (MSE 20.37) selected only 4 predictors: Age, Abdomen, Forearm, Wrist

Elastic Net achieved the best test MSE (20.28) with the same 4 predictors

Abdomen circumference is the single most important predictor of body fat %